11. Comments on Final Projects

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Category	Number of Students
Factor Model	2
Beta Hedging	1
Sentiment	1
Sports Betting	2
Spot-Future	1
Position Control/Regime Switching	2
RWA Arbitrage	1
Limit Order Book	1
Still Working	2

Strategy 1: Trading on Factor Models (2 students)

Targeting Market	Cross-region equity markets across developed and emerging economies, including the United States, Europe, China, Japan, and selected emerging markets (India, Korea, Brazil, South Africa).
Phenomena Description	Regional equity markets often move together at a broad level, yet leadership typically rotates over medium-term horizons. These relative performance differences persist due to slow-moving capital flows, differences in monetary and fiscal policy, growth and inflation dynamics, and heterogeneous exposure to global risk appetite and currency conditions.
Theoretical Framework	Macro-finance and asset pricing concepts, including CAPM-style systematic risk, global risk-on/risk-off regimes, behavioral explanations for momentum persistence, and limits to arbitrage arising from trading frictions, institutional inertia, and policy uncertainty.
Theoretical Hypothesis	• Relative momentum across regional equity markets persists over 6–12 month horizons because capital reallocates gradually rather than instantaneously. • The persistence of relative momentum is regime-dependent and weakens during financial crises or abrupt policy shocks when correlations spike and leadership reverses. • A small set of macroeconomic and bond-market indicators can identify regimes in which cross-region trends are more likely to persist.
Phenomena Interpretation	Observed return differentials are interpreted as macro-driven divergence combined with gradual investor response and benchmark rebalancing, rather than pure mispricing. Regions benefiting from favorable growth, inflation, policy, or external balance conditions attract sustained capital inflows until the macro regime shifts.
Trading Opportunity	Exploit tradable dispersion in regional equity leadership by taking long exposure to strong-performing regions and short exposure to weak-performing regions, aiming to generate returns from relative outperformance while reducing sensitivity to overall market direction.

Table 1. Core Logic Design of Trading Strategies

It is a long-short global macro strategy based on factor models.

- **Phenomena to problems.**

1. How to select the effective factors?
2. What are your risk exposures? Is your strategy market neutral? You need to define which factor is the primary one for control. e.g., for a long-short strategy, the dominant factor to manage is usually the net market risk exposure.
3. This strategy is primarily momentum-driven at the global level. When to open/exit a position? What are the rebalancing frequency and costs?
4. How to construct an optimzied portfolio based on constraint factor risk exposures?
5. Is there any way to reduce the number of assets in the portfolio? (Minimuze costs)
6. What’s the best long/short ratio?
7. Any opportunities to engineer the “portable” alpha?

Strategy 2: Trading on Co-movements, Logic Check

Targeting Market	Mid-cap tech / industrial equities (and closely related such as key suppliers, commodity proxies, or major "price-mover" firms).
Phenomena Description	After a shock to a *primary mover*, economically dependent *secondary tickers* adjust with a delay. Alpha comes from the transient mispricing during this lead–lag window.
Analytical Framework	(1) Build a dynamic dependency graph using LLM extraction over 10-K risk factors / major customer language, trade journals, local news, and earnings-call transcripts. (2) For each candidate, estimate cointegration and monitor the spread as a stationary process.
Phenomena Interpretation	x_t = log price of the *primary mover* and y_t log price of the *secondary tickers*. The spread $y_t - \beta x_t = \epsilon_t$ is stationary (cointegration). Large deviations represent temporary dislocation caused by asynchronous information diffusion across linked firms.

Table 1. Core Logic Design of Trading Strategies

- "Mid-cap" is an assumption. It may not be necessary to make such assumption so early. We should test it during backtests.
- "Economically dependency", "lead-lag window" and "Alpha" are the core to the strategy. It is clearly stated.
- **Decompose the phenomena into problems to be solved.**

1. How to identify the "dependent" stocks?
2. How to make trading decisions based on their dependency?

- **Matching problems to models.**

1. Dependency-> LLM-> Pairs
2. Temporary dislocation -> Statistical tool ->trading signals

Strategy 2: Trading on Co-movements, Model Check

Data Requirement	- news feeds - trade journals - earnings call transcripts (for LLM graph building) . - 1-minute or 5-minute Open, High, Low, Close, Volume data for the selected pairs to calculate spreads and execution timing.
Quantitative Model	- To estimate the equilibrium relationship $y_t - \beta x_t = \mu$ - To normalize the spread for trading signals $Z = \frac{\epsilon_t - \mu}{\sigma}$
Strategy Type	Statistical Arbitrage / Pairs Trading (Mean Reversion) enhanced by Event-Driven signals
Trading Decision Triggers	- Entry: Z-score crosses threshold (e.g., +/- 2.0). - Exit: Z-score reverts to mean (0.0) or crosses stop-loss (+/- 3.5). - Filter: LLM "Linkage Confidence Score" > 80% (only trade pairs where the supply chain link is verified).

Table 2. Design of Algorithmic Models

- **“Matching problems to models.”**

- “Economic link”-> 10-K, news, etc.-> LLM
- “Statistical link” -> Correlation, Stationarity, The Granger, variable-lag Granger, and Toda-Yamamoto causality tests
- “Short-term” reaction delay -> spread “mean-reversion” -> Z-score
- “Alpha” -> linear regression

- A model built on solid logic doesn’t need to be complex. Use a simple model as the benchmark to evaluate whether more advanced models or methods truly offer improved performance.

Strategy 2: Trading on Co-movements, Same Phenomena under Different Framework

Theoretical Reasoning	When the primary mover re-prices immediately after a dependency-relevant event, the dependent ticker underreacts (or overreacts) briefly. As broader participants process the linkage, the dependency tickers catches up and spread reverts toward its mean. Better graph quality = cleaner pair selection = fewer "false link" trades.
Trading Opportunity	Mean reversion in the spread driven by real economic linkage + short-term reaction delay.
Trading Operations	**Universe filter:** mid-cap tech/industrial names with strong dependency on a commodity or key firm (for example, huge revenue dependency, when identifiable). **Signal:** compute Z-score of the spread.

Table 1. Core Logic Design of Trading Strategies

- When you applied different analytical framework, you could have different reasoning for the same phenomena.
- Don't give up a idea too soon. If the logic is solid, sometime you could change the analytical framework.
- Systemic Risk Framework: There is a spillover effect between systemically important stocks and non-systemic stocks. The impact of repricing is also asymmetric, with 'central' stocks exerting stronger influence than their 'non-central' counterparts.
- Under Systemic Risk Framework. the model could be Co-VaR, Tail-dependence, etc.

Strategy 3: Trading on Sentiment Analysis, Logic Check

Core Logic Design of Trading Strategies	
Targeting Market	~~Popular cryptocurrencies like BTC & ETH. (Ones with more exposures to the public and social media)~~ After reading some papers, the BTC & ETH markets tend to be less affected by market sentiment, as most participants are informed investors (Baur and Dimpfl, 2018). Therefore, we choose SOL as our target since it has a relatively large market capitalization and a relatively lower price per coin (which is more accessible for individual investors. Not sure if the latter one counts. May be "Unit Price Bias"?).
Phenomena Description	The crypto market is an adaptive market that affected by investors' behavior, market psychology, and availability of new information. Increased correlation among crypto assets and participants without fundamental knowledge.
Analytical Framework	Theoretical: Adaptive Market Hypothesis (AMH) & Behavioral Finance. Methodological: Social Media Analysis (NLP) to construct a "Behavioral Index" similar to VIX.
Phenomena Interpretation	Since the crypto market is sensitive, speculative, and increasingly correlated, high levels of social media activity amplify behavioral biases such as overconfidence and anchoring, leading to herd mentality among investors. This collective behavior causes cryptocurrency prices to temporarily deviate from their fundamental values, resulting in market anomalies.
Trading Opportunity	Identify "Panic" (Oversold) or "Euphoria" (Overbought) moments via the Behavioral Index.
Trading Operations	1. Calculate the index on a certain rolling window as benchmarks. 2. Entry: Buy when indicator suggests "Panic" / Sell when "Euphoria" 3. Exit: Profit before the market reverts to the mean
Financial Instrument(s)	Spot SOL or corresponding derivatives such as perpetual futures or futures contracts to allow both long and short positions.

Table 1. Core Logic Design of Trading Strategies

- It is good to keep recording your thoughts. The tables are actually your notebook for strategy development rather than assignments.

  There is some confusion between the theoretical and empirical tracks of strategy design. When developing a theoretical framework, you should begin with underlying assumptions rather than observed phenomena.

  Decompose the phenomena into problems to be solved.

  - How to model or measure sentiment?
  - How to measure "Sentiment Momentum"?
  - How to measure "Sentiment Decay Effect"?
  - How does sentiment affact crypto prices? (Sentiment -> Behavior -> Price)

  Matching problems to models.

  - Sentiment Analysis-> NLP -> Sentiment Index-> Behavioral Index -> Asset Price
  - Sentiment "Mean-Reversion"-> Buy "Panic"/Sell "Euphoria"

Strategy 3: Trading on Sentiment Analysis, Model Check

Design of Algorithmic Models	
Data Requirement	5-minute (or 15-minute) OHLCV data for SOL, Trading volume data, Social media text data (e.g., Twitter / Reddit) filtered by keywords ("SOL", "Solana", "$SOL"), Timestamped sentiment scores derived from text data
Quantitative Model	$BI_t = Z(S_t) + Z(A_t) + Z(V_t)$
Strategy Type	Mean-reversion driven by market sentiment, event-driven which affected by volatility changes during "Panic" (Oversold) or "Euphoria" (Overbought) moments
Trading Decision Triggers	Threshold-based signals using Behavioral Index: • Long entry when $BI_t < -\theta$ (panic); • Short entry when $BI_t > \theta$ (euphoria); • Exit when BI_t reverts toward zero or after a fixed holding period.
Parameters* (definition and possible ranges)	S_t = average sentiment score in interval t Range: [-1, 1] $A_t = \log(1 + \text{number of posts mentioning SOL in } t)$ $V_t = \log(\text{trading volume}_t)$

Table 2. Design of Algorithmic Models

- There is some confusion between variables and parameters. Parameters are usually time window, threshold, range, etc.

- **Matching problems to models.**

1. Price mean-reversion driven by sentiment -> "volatility" change -> sentiment "moments"

2. The model assumes BI=f(S, A, V). Check the relations among S, A, V before modeling. They could be correlated, colineared, casual, etc.

Strategy 4: Trading on Prediction Market, Logic Check (2 students)

Targeting Market	Polymarket NBA game outcome markets
Phenomena Description	Prediction markets often misprice NBA games relative to true win probabilities. Market prices reflect public sentiment and simple metrics rather than comprehensive statistical models. Retail-heavy markets exhibit recency bias (overweighting recent performance), home team bias, and overreaction to incomplete information like single-game injuries or hot streaks.
Theoretical Framework	Efficient Market Hypothesis suggests markets aggregate all available information, but sports prediction markets have structural inefficiencies due to retail-heavy participation, recency bias, and limited quantitative modeling by participants. Unlike financial markets dominated by institutional capital, prediction markets remain fragmented with significant information asymmetries.
Analytical Framework	Machine learning model processes historical NBA data including team statistics (offensive/defensive efficiency, pace, true shooting percentage), player performance metrics (usage rate, PER, plus-minus), injury reports, matchup histories, home/away splits, rest advantages (back-to-backs, days of rest), and schedule context. Model generates pre-game win probabilities using ensemble methods (XGBoost, Random Forest, Neural Networks). Compare model probabilities to market-implied probabilities to identify mispriced games where edge exceeds threshold.
Theoretical Hypothesis	Markets systematically misprice certain game types creating exploitable edges: (1) Heavy underdogs in specific matchup contexts are overvalued due to public fade, (2) Back-to-back situations are underpriced for rested teams, (3) Rest advantages (3+ days vs 1 day) create measurable edges not fully priced in, (4) Home court advantage is overweighted in public perception, (5) Star player absence is overreacted to without context of team depth.

Table 1. Core Logic Design of Trading Strategies

- It is a good point that sports betting activities are not “efficient”.
- Prediction market is new and it allows certain events to become tradable products.

- **Decompose the phenomena into problems to be solved.**

1. How to model an NBA team?
2. How does the team data relate to its performance?
3. How does the opponent affact the team’s performamce?

- **Matching problems to models.**

1. NBA team-> team statistics-> Ensemble ML algorithms -> pre-Game probability
2. Mispriced market probability -> model implied probability

Strategy 4: Trading on Prediction Market, Logic Check

Targeting Market	**Polymarket NBA game outcome markets**
Phenomena Interpretation	When model probability significantly diverges from market price (e.g., model shows 60% vs market 50%), an edge exists. The magnitude of divergence indicates sizing opportunity via Kelly Criterion. Edge = Model Probability - Market Probability. Positions are sized proportionally to edge magnitude and confidence level (based on model uncertainty estimates).
Theoretical Reasoning	Prediction markets allow continuous trading unlike traditional sportsbooks, enabling dynamic risk management. Zero commissions on Polymarket preserve edges that would erode on traditional platforms. Real-time position adjustment captures value from game flow volatility that static bets cannot exploit. Live trading enables asymmetric payoff structures: cap downside through stop losses, maximize upside through probability-driven profit targets. Market microstructure allows exploitation of momentum and mean-reversion patterns during live games.
Trading Opportunity	Enter positions when model probability exceeds market price by threshold (5%+ edge minimum). Size positions using Kelly Criterion based on edge magnitude and model confidence. Target 20-30 games per week during NBA season (October-April). Focus on games with: (1) Rest mismatches (3+ day differential), (2) Back-to-back scenarios, (3) Matchup-specific edges (pace, defensive rating mismatches), (4) Injury situations with depth advantages. Avoid nationally televised games and playoff games where market efficiency increases due to higher volume and sharper participants.
Trading Operations	**Pre-game Phase:** Bot scans upcoming games 2-4 hours before tip-off when lines stabilize. ML model generates win probabilities for both teams. Compare model odds vs market odds to identify edges exceeding 5% threshold. Filter for games meeting volume requirements (minimum $10K liquidity). Calculate position size using fractional Kelly (0.25-0.5 Kelly for risk management). Execute entry orders via API. **Live Trading Phase:** WebSocket monitors real-time game state (score, time remaining, possession, momentum indicators). Execute stop losses if position moves against pre-determined threshold (typically 15-20% probability shift or position value decline >40%). Execute take profits when favorable momentum creates exit opportunity: (1) Underdog positions: Exit on leads of 8+ points after Q1, 12+ points after Q2, (2) Favorite positions: Exit when lead reaches 2x expected margin or when position value increases 80%+. Continuous probability updating during game adjusts thresholds dynamically.
Financial Instrument(s)	Binary outcome contracts on Polymarket (Team A wins / Team B wins). Each contract pays $1.00 if correct, $0.00 if wrong. Trading occurs at prices between $0.01-$0.99 representing implied probability.

Table 1. Core Logic Design of Trading Strategies

1. Kelly criterion on a binary game. Sharpe ratio is also related to Kelly criterion under EMH.
2. There are multiple trading rules and scenarios pre-defined. During the backtests, we need to estimate each of them to see if it actually works or not.

Strategy 4: Trading on Prediction Market, Model Check

Data Requirement	**Historical Game Data:** Team-level metrics (Offensive Rating, Defensive Rating, Pace, True Shooting %, Effective FG%, Turnover Rate, Rebound Rate, Free Throw Rate). Player-level metrics (PER, Usage Rate, Win Shares, Plus-Minus, On/Off differentials). Game context (Home/Away, Days Rest, Back-to-back indicator, Travel distance, Altitude). **Live Game Data (1-min intervals):** Current score, Time remaining, Possession count, Lead changes, Largest lead, Momentum indicators (scoring runs, shooting percentages over last 5 minutes). **Injury/Roster Data:** Starting lineups, injury reports, minutes restrictions, load management indicators. **Market Data:** Real-time Polymarket prices, order book depth, volume, recent price movements.
Quantitative Model	**Pre-game Model:** Ensemble approach combining XGBoost (gradient boosting for non-linear relationships), Random Forest (robustness to outliers), and Neural Network (deep feature extraction). Features engineered from rolling averages (5-game, 10-game, 20-game windows) to capture form. Model outputs: Win probability for each team, Confidence interval (uncertainty quantification), Feature importance scores. **Live Game Model:** Dynamic Bayesian updating framework that adjusts pre-game probabilities based on in-game developments. Incorporates score differential, time remaining, possession count, and momentum metrics. Win probability updated every minute during live games.
Strategy Type	Hybrid approach combining statistical arbitrage (pre-game edge identification) and momentum/mean-reversion (live trading adjustments). Statistical arbitrage component exploits pricing inefficiencies identified by ML model. Momentum component captures value from early leads and hot shooting. Mean-reversion component fades overreactions to small sample variance.

Table 2. Design of Algorithmic Models

- **Matching problems to models.**

1. The design of the strategy is clear：
Pre-game, In-Game models can capture both “fundamentals” and “intraday price movements”.

2. Using the ML for solving such problem is a good choice. But instead of predicting one team, I’d suggest to take the opponents into consideration since the performance is actually based on competitions between 2 teams.

3. Since betting on a game is similar to pricing a binary option, an option-liked portfolio can also be appied to construct this type of strategy. (e.g., straddle, short butterfly, etc.)

Strategy 5: Trading on Spot-Future Co-movment

Core Logic Design of Trading Strategies		
Targeting Market	The BTC futures basis market during demand dislocations — specifically, the temporary mispricings that emerge when spot demand surges and the term structure inverts into backwardation.	
Phenomena Description ☒	Theoretical Framework ☐	The futures curve goes into backwardation for demand reasons.
Analytical Framework ☒	Theoretical Hypothesis ☐	F=S×e^(r+u-y)T, volatility-based signals (5-day MA percentile, or peaks in the volatility of volatility)
Phenomena Interpretation ☒	Theoretical Reasoning ☐	If there's a situation that can you make money on the demand of crypto.
Trading Opportunity	The basis arbitrage strategy logic. If the basis is negative, then I will short the BTC spot, and long the BTC futures.	
Trading Operations	When the basis turn to negative during a top-percentile volatility regime, go long BTC futures and short BTC spot, then unwind both positions when the basis reverts toward zero or its theoretical fair value.	
Financial Instrument(s)	BTC monthly future contracts and monthly BTC underlying data and S&P500 Index	

Table 1. Core Logic Design of Trading Strategies

1. It is a typical spot-future pairs trading strategy.

2. The phenomena is not detailed enough.

- **Matching problems to models.**

1. How to measure the "demand" of BTC? Data related to mining and transactions from the blockchain should be included for estimate the "demand".

2. How does the "demand" affact BTC spot price and future price? (May be factor models)

3. How to decide when to open a position?

Strategy 6: Trading on Position-Control/Regime-Switching based strategy (2 students)

Component	Description
Targeting Market	Broad financial markets with multiple correlated assets, such as the crypto market or equity market, where systemic risk and correlation changes are more pronounced during stressed periods.
Theoretical Framework	This strategy is based on **Network Theory** and **Chaos Theory**, which view financial markets as complex and interconnected systems. Network theory explains how correlations form risk transmission channels, while chaos theory explains how markets can become unstable and highly sensitive to small shocks.
Theoretical Hypothesis	When asset correlations rise significantly and the market enters an unstable period, risk spreads quickly across the system and diversification becomes ineffective. Exiting long positions during these periods can reduce large losses and improve risk control.
Theoretical Reasoning	In normal periods, assets are loosely connected and shocks remain localized. As correlations increase, the market network becomes denser, allowing losses to spread out across assets. At the same time, chaotic dynamics increase unpredictability, making price movements unreliable. Reducing exposure during these highly connected and unstable periods is therefore theoretically justified.
Trading Opportunity	The opportunity lies in **avoiding major drawdowns** during periods of systemic risk and instability, rather than attempting to predict short-term price movements or market timing.
Trading Operations	Monitor overall market connectedness and signs of instability. Exit long positions when the market shows strong correlation and chaotic behavior. Re-enter positions gradually after correlations and instability decrease in post-crash periods.
Financial Instrument(s)	Broad market instruments such as spot assets, index products, ETFs, inverse ETFs, or volatility-related instruments, depending on the specific market being traded.

Table 1. Core Logic Design of Trading Strategies

1. It is a long-only strategy with dynamic position control.

2. The statement of theoretical hypothesis is more likely to be a phenomena.

- **Matching problems to models.**

1. How to measure the systemic risk and system instability? The correlation matrix can be considered as a fully connected network with edge weights are correlations. There are more types of complex networks such as minimum spanning tree, small-world, scale-free, random network, etc. They may be used for describe different features.

2. How do realized volatility and implied volatility related to market crash?

3. How to optimize your position level based on systemic risk?

4. Which method is better, hedging or cashing-out?

Strategy 7: Trading on RWA-Underlying Arbitrage

Table 1: Core Logic Design of Trading Strategies	
Targeting Market	RWA-related crypto tokens with exposure to real-world economic activity
Theoretical Framework	RWA prices often deviate from real-world supply–demand fundamentals in the short term Price reactions to macro events, regulation, or supply chain shocks are delayed or uneven across markets Chain-based markets and real-economy signals adjust at different speeds Relative value framework Supply-demand imbalance theory Market segmentation between on-chain markets and real-world economic systems
Theoretical Hypothesis	Relative price spread analysis between: • RWA price movements • Supply chain conditions • Stablecoin funding rates / liquidity conditions RWA prices maintain a stable long-term relationship with real-world fundamentals Temporary deviations arise from information lags, regulatory friction, or liquidity shocks These deviations are expected to mean-revert over time → Standardized price spread (e.g. z-score or normalized deviation from historical mean)
Theoretical Reasoning	• Price discrepancies reflect lags in the price discovery process • On-chain markets react faster to liquidity and sentiment • Real-world signals adjust more slowly • This speed mismatch creates short-term inefficiencies
Trading Opportunity	• Enter positions when the relative price spread exceeds historical norms (e.g. spread exceeds ±2 standard deviations) • Capture returns from convergence toward long-term equilibrium

Trading Operations	• Establish positions when observed spreads exceed predefined thresholds
	• Reduce or close positions when:
	  o Spreads converge toward zero
	  o Upstream supply chain or financing signals reverse
	• Apply conservative position sizing during high uncertainty
	• Set upper limits on total risk exposure
	• Position size scaled with spread magnitude but capped at portfolio level

Financial Instrument(s)	• RWA-related tokens • Stablecoins • RWA-linked ETFs or derivatives

Table 1. Core Logic Design of Trading Strategies

1. It is a spot-derivative arbitraging strategy.

RWA-related assets are connected by economic fundamentals and often share the same underlying. A core assumption of this strategy is that any price gap between two contracts is “temporary” rather than a “permanent” depeg.

2. Example: TSLAx (tokenized Tesla stock) and TSLA (regular stock); XAUUSD (Spot Gold), GLD (gold ETF), UGL (2x Long Gold Leveraged ETF),etc.

- **Matching problems to models.**

1. How to select the pairs for trading? If two contracts are too similar, the price difference may not large enough to cover your trading costs.

2. It is betting a mean-reverting process of the spread. When to open/exit a position?

Strategy 8: Trading on Limit Order Book, Logic Check

Targeting Market	Electronic markets with a continuous limit order book and sufficient liquidity (BTC / ETH / ADA spot markets).
Phenomena Description	Prices fluctuate frequently in the short term, and there is usually a gap between the best buying and selling price. Order depth and cancellation behavior change over time.
Theoretical Framework	Market microstructure and limit order book mechanics: liquidity providers earn spread while taking inventory and adverse selection risk.
Analytical Framework	Use historical order book fields: midpoint, spread, bid/ask notional volumes, cancellation ratios, and short-term volatility to guide order placement.
Theoretical Hypothesis	Providing liquidity through limit orders near the midpoint can generate positive expected returns when volatility is moderate and liquidity is stable.
Phenomena Interpretation	Short-term price oscillations and temporary order imbalance create repeated opportunities to buy slightly lower and sell slightly higher.
Theoretical Reasoning	Traders who want immediate execution cross the spread; liquidity providers earn the spread as compensation for taking execution and inventory risk.
Trading Opportunities	Capture small profits repeatedly through spread execution while avoiding unstable periods detected by high volatility or high cancellation activity.
Trading Operations	Place limit buy and sell orders around the midpoint, update distance using measured volatility, control position size, and pause trading when spread or cancellation ratio exceeds a threshold.
Financial Instrument	Any asset traded through a limit order book (cryptocurrencies, stocks, or futures).

Table 1. Core Logic Design of Trading Strategies

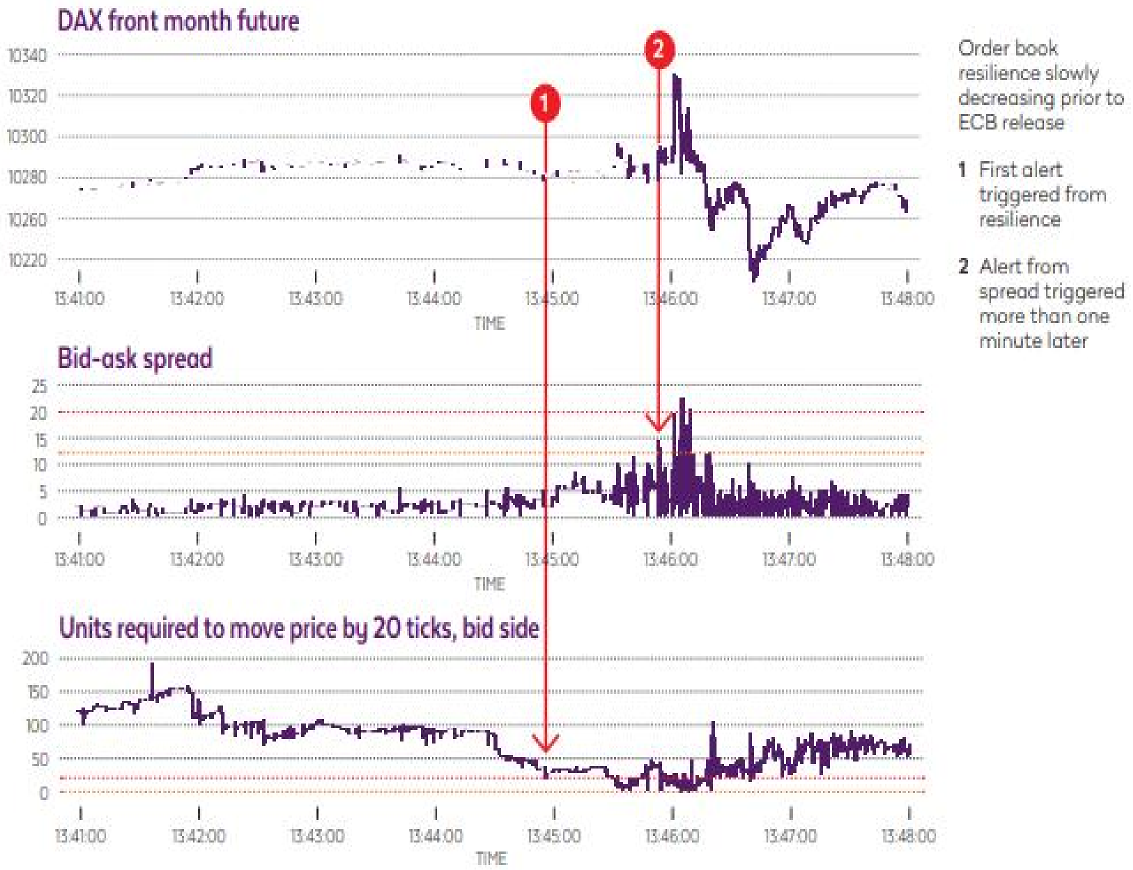
1. There's some confusion about theoretical and empirical frameworks. Usually you only need to pick one track. The theoretical framework shall start with assumptions such as random walk hypothesis, exponentially distributed limit orders, new order arrival follows Poisson distribution, etc.
2. This strategy primarily focuses on short-term and ultra–short-term price mean-reversion patterns, avoiding high volatility and high cancellation rate of order books.
3. There are more ways to trading on LOBs. For example, you could monitor the LOBs from spot market, future market, and option market at the same time. You could hedge your inventory or trade on option combos, etc.
4. Take time zone into considerations for cryptos.

- **Matching problems to models.**

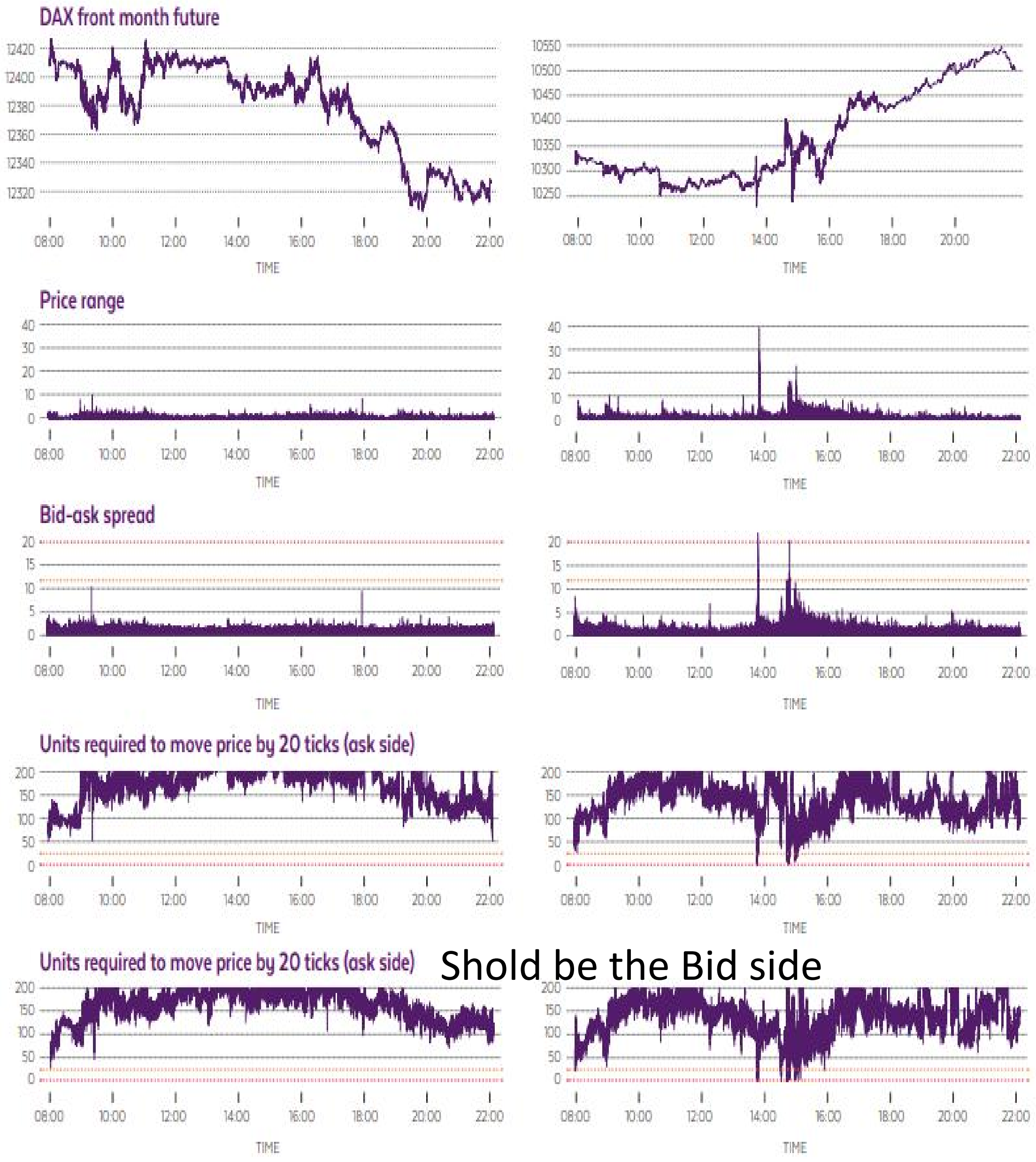
1. How to model the short-term price mean-reversion pattern?
2. How to avoid trading the trends?
3. How to predict/recognize high-volatility/price-jump events?
4. How to detect high cancellation rate of LOB?

Strategy 8: Trading on Limit Order Book, Model Example

Figure 5: Alert comparison on 22 January 2015



Comparison of the sensitivities of alerts based on bid-ask spread and order book resilience around the price spike – with subsequent decline – around the ECB interest rate announcement on 22 January 2015.



Shold be the Bid side

source: Dr. Stefan Teis & Georg Gross

Thank You

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